

Checking Validity of an argument
with an indirect truth table

Indirect Truth Table

Indirect truth tables provide a faster way for checking validity of arguments.

- ◆ The idea is to check whether the argument is possible to have all true premises and a false conclusion by making assumptions.
- ◆ If it is possible for the argument to have all true premises and a false conclusion, then the argument is **invalid**. Otherwise, **valid**.

Indirect Truth Table

We need to first **assume the conclusion is false and all premises are true**(which means it is invalid).

Then we have to see if there is any **contradiction** in the computation.

➤ If there is a contradiction, then the assumption is false ,
i.e. the argument is **valid**.

➤ If there is **no** contradiction, then there is a case that the conclusion is false and the premises are true, i.e. the argument is **invalid**.

Case 1 : $\sim(P \vee R)/Q \equiv \sim P//Q \cdot \sim R$

- If there is a contradiction, then the assumption is false , i.e. the argument is **valid**.
- If there is **no** contradiction, then there is a case that the conclusion is false and the premises are true, i.e. the argument is **invalid**.
- In the indirect truth table of this argument ,there is a contradiction.
- **We need to indicate the contradiction by circling it.**
- It is **valid**!!!!

$\sim$	(P	$\vee$	R)	/	Q	$\equiv$	$\sim$	P	//	Q	$\cdot$	$\sim$	R
T	F	F	F		F	T	F	T		F	F	T	F
①	③	②	③		⑦	①	⑧	⑨		⑥	①	⑤	④

Case 2: $P \supset S / (S \cdot \sim Q) \vee P // P \vee Q$

- We find no contradiction.
- It is an **invalid** argument since we can find a row (a logical possibility) with all true premises and a false conclusion.

P	$\supset$	S	/	(S	$\cdot$	$\sim$	Q)	$\vee$	P	//	P	$\vee$	Q
F	T	T		T	T	T	F	T	F		F	F	F
3	1	7		6	5	6	4	1	3		2	1	2

Exercise : $A \cdot E / (A \vee D) \equiv \sim B // D \supset B$

Invalid

$A \cdot E / (A \vee D) \equiv \sim B // D \supset B$													
T	T	T		T	T	T	T	T	F		T	F	F
3	1	3		7	6	8	1	5	4		2	1	2

Indirect Truth Table

□ Sometimes we need to make further assumption if the truth value of certain letters are not fixed by previous steps.

P	$\supset$	Q	/	Q	$\supset$	P	//	P	$\equiv$	Q
T	T	F		F	T	T		T	F	F
F	T	T		T	T	F		F	F	T

□ If no contradiction is found, then the argument is invalid, you can stop further computation.

□ If a contradiction has been found, you **CANNOT** stop the computation. **You need to exhaust every logical possibility.**

Case 3: $P \supset Q / Q \supset P / P \equiv Q$

- Again assume that all premises are true and the conclusion is false.
- However, we cannot determine the truth value of “P” and “Q” without further assumptions
- “ $P \supset Q$ ” is true iff
 - (a) both “P” and “Q” are true ,or
 - (b) “P” is false and “Q” is true, or
 - (c) “P” is false and “Q” is false.
- For “ $Q \supset P$ ”, **there are three possibilities for it to be true.**
- For “ $P \equiv Q$ ”, **there are two possibilities for it to be false.**
- Therefore, we have to make further assumptions.

Case 3: $P \supset Q / Q \supset P / P \equiv Q$

- We will start from “ $P \equiv Q$ ” since there are only 2 cases which “ $P \equiv Q$ ” is false instead of 3 cases to be true for the premises.
- “ $P \equiv Q$ ” is false iff
- (a) “ P ” is true and “ Q ” is false; or (b) “ P ” is false and “ Q ” is true;
- We will **assume “ P ” is true and “ Q ” is false first.**
- **If there is no contradiction, we can stop since we already found a line with all true premises and a false conclusion.**
- **But if there is a contradiction, we need to continue to look at other cases until we exhaust all the possibilities.**

P $\supset$ Q /			Q $\supset$ P //			P $\equiv$ Q		
T			T			T	F	F
1			1			2	1	2
						F	F	T

Case 3: $P \supset Q / Q \supset P // P \equiv Q$

- Since there is a contradiction in all cases, i.e. we cannot find a row (logical possibility) that all premises are true and the conclusion is false.
- Therefore, the argument is **valid**

P	$\supset$	Q	/	Q	$\supset$	P	//	P	$\equiv$	Q
T	T	F		F	T	T		T	F	F
3	1	4		4	1	3		2	1	2
F	T	T		T	T	F		F	F	T
3	1	4		4	1	3		2	1	2

Example

Whether the following argument is valid?

$(P \equiv Q) \supset S$ / $\sim S \cdot P$ / $\therefore P \vee Q$

Valid

$(P$	$\equiv$	$Q)$	$\supset$	S	/	$\sim$	S	$\cdot$	P	//	P	$\vee$	Q
F	T	F	T	F		T	F	T	T		F	F	F
5	8	7	1	6		2	4	1	2		3	1	3